**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 067 · Two-Dimensional Arrays

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. How can a loop total an array's values?

Q2. What is an embedded system designed for?
A. Any desktop application
B. Only browsing websites
C. A specific task
D. Only storing backups

Q3. “Adding active users always increases the bandwidth available to each user.” Correct this. Explain why.

Q4. A school needs central accounts and shared files. Three club laptops share files directly. Correct both claims. Explain why. (a) Every client must be a dedicated server. (b) Peer-to-peer needs a central server.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

# I Do · Read new knowledge

### Key vocabulary

2D array — Stores data in a fixed-size grid accessed by row and column indices.

Row index — The first index identifies a particular row; indexing starts at zero.

Column index — The second index identifies the column within the chosen row.

Grid assignment — A value is set or retrieved using both its row and column coordinates.

Syntax — The rules governing valid programming statements.

Test data — An input used to check expected results.

### New learning

2D array: Stores data in a fixed-size grid accessed by row and column indices. Apply this in a theatre seats.

Row index: The first index identifies a particular row; indexing starts at zero. Apply this in a school seating plan.

Column index: The second index identifies the column within the chosen row. Apply this in a game board.

Grid assignment: A value is set or retrieved using both its row and column coordinates. Apply this in a class timetable.

### Worked example

A theatre seats is being developed. Explain how 2d array and row index are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1].

3. Row index [1]; The first index identifies a particular row; indexing starts at zero. [1].

4. Two linked points for each concept.

# We Do · Practise with guidance

### Guided 1

A school seating plan uses 2d array. Explain how it helps. [2 marks]

Worked method: 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1].







Q3. 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1]. Column index [1]; The second index identifies the column within the chosen row. [1]. Grid assignment [1]; A value is set or retrieved using both its row and column coordinates. [1].

### Exit Ticket · Model answer

True

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Set a total to 0. Loop through the array, adding each value once.

Q2. C — A specific task

Q3. Users may compete for finite bandwidth, reducing the data rate available to each.

Q4. (a) A server manages shared services and resources for clients. (b) Peers share directly and can act as clients and servers. Central control suits a school; peer-to-peer can cost less for a small club but is harder to manage as it grows.

### I Do / We Do pages · Guided practice

Q1. 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1].

Q2. Row index [1]; The first index identifies a particular row; indexing starts at zero. [1].

Q3. Column index [1]; The second index identifies the column within the chosen row. [1]. Grid assignment [1]; A value is set or retrieved using both its row and column coordinates. [1].

### You Do pages · Code First

```
array desk[2,2]
desk[0,0] = "PC1"
desk[1,0] = "PC2"
print(desk[1,0])
```

Expected result: PC2

### You Do pages · Independent questions

Q1. 2D array [1]; Column index [1].

Q2. Row index [1]; The first index identifies a particular row; indexing starts at zero. [1]. Column index [1]; The second index identifies the column within the chosen row. [1].







### Guided 2

Explain row index in a game board. [2 marks]

Worked method: Row index [1]; The first index identifies a particular row; indexing starts at zero. [1].

### Guided 3

A class timetable uses column index and grid assignment. Explain both. [4 marks]

Worked method: Column index [1]; The second index identifies the column within the chosen row. [1]. Grid assignment [1]; A value is set or retrieved using both its row and column coordinates. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Print the label at row 1, column 0.

```
array desk[2,2]
desk[0,0] = "PC1"
desk[1,0] = "PC2"
```

write

### Question 1

State TWO relevant techniques from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

### Question 2

Explain how row index and column index help with a museum room matrix. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

### Question 3

A programmer designs a parking spaces. Explain why 2d array, column index and grid assignment matter. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.



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**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Exit Ticket · Answer before checking

A two-dimensional array stores data in a grid accessed by row and column indexes.